Trial Schedules

Table 1 Flow Chart for Efficacy and Safety Evaluation, for Screening, IMP Administration + Post-Treatment Follow-up (Week -6 to Week 52)

Trial Period	**Screening**	**IMP Administration**			**Post-treatment Follow-up**						**Additional Visits** [a]
Weeks/Day	W-6	D0 pre IMP [b]	D0 0h	D0 post IMP	D1	D3	W1 to 12 [b, c] (1 visit/W)	W14 to 26 [b, c] (every 2nd W)	W31 to 48 [b, c] (1 visit/month)	W52 [b, c]	
Visit Window (Days)	0	0	0	0	0	±1	±2	±3	±5	±5	
Informed consent	x										
In-/exclusion criteria	x	x									
Medical history	x	x									
Collection of historical bleeding and factor IX use data	x										
24-hr stay at clinic		x----------x									
Administration of IMP			x								
Discharge and hand-out of subject treatment card [d]					x						
Concomitant medication	Continuous										(x) [a]
Vital signs (blood pressure, pulse, body temperature)	x	x		x [h]	x [h]						(x) [a]
Physical examination	x	x					x	x	x	x	(x) [a]
Height	x										
Body weight	x										(x) [a]
Blood & semen sampling (see Table 2)	x	x		x	x	x	x	x	x	x	(x) [a]
Subject hand-out e-diary device and e-diary instruction [e]	x										
Recording of e-diary data by subject (bleeding episodes, factor IX replacement therapy)	Continuous										
Review of e-diary data by Investigator/study nurse	Continuous										(x) [a]

Trial Period	Screening	IMP Administration			Post-treatment Follow-up						Additional Visits [a]
Weeks/Day	W-6	D0 pre IMP [b]	D0 0h	D0 post IMP	D1	D3	W1 to 12 [b, c] (1 visit/W)	W14 to 26 [b, c] (every 2nd W)	W31 to 48 [b, c] (1 visit/month)	W52 [b, c]	
Visit Window (Days)	0	0	0	0	0	±1	±2	±3	±5	±5	
Collection of e-diary device										x	
Subjects provided with paper diaries and paper diary instruction for Long-term Follow-up Phase										x	
Adjudication of bleeding episode(s) by Investigator[f]	Continuous										(x) [a]
CCI		x						x (w26)		x	
CCI		x						x (w26)		x	
CCI		x						x (w26)		x	
CCI		x						x (w26)		x	
CCI		x						x (w26)		x	
CCI		x								x	
Adverse events	Continuous										(x) [a]

CCI IMP: investigational medicinal product; ; W: week.

a. An additional visit can be performed at any time for the purpose of conduct of one or more procedures listed in the column "Additional Visits" according to the choice of the Investigator. Hence, "(x)" refers to a procedure that can be performed, if judged relevant by the Investigator.
b. For subjects on routine prophylactic factor IX replacement therapy, these visits should take place on the day that routine prophylactic factor IX replacement treatment is planned to be administered. At these visits, blood sampling will then take place prior to administration of prophylactic factor IX replacement therapy. If a subject uses additional on-demand factor IX replacement treatment, his upcoming study visit may need to be re-scheduled (to the extent the visit window allows) so that the visit takes place at the time the subject resumes his routine prophylaxis schedule. For subjects only using on-demand factor IX replacement therapy (only applicable after AMT-061 administration), his upcoming study visit may need to be re-scheduled (to the extent the visit window allows) so that the visit does not take place within 10 days of any on-demand factor IX product use.
c. Each visit is scheduled in relation to date of IMP administration, not in relation to the date of its previous visit.
d. When all assessments at 24 hours after IMP administration are performed, the subject treatment card will be handed out and the subject may leave the clinic.
e. Instruction can be repeated at any visit as deemed necessary by the Investigator or study nurse.
f. The Investigator or designee will assess each bleeding episode, by describing the circumstances as well as the nature of the reported bleed in a bleed specific narrative as soon as possible but at least within 72 hours after it has been reported by the subject.
g. It is recommended that the CCI (CCI) are completed by the subject, before he is seen by the Investigator and/or study nurse for interview and before other (trial specific) assessments are performed, as well as before the administration of IMP, in order for the answers not to be influenced by the information given by the physician, by the administration of IMP, or by early side-effects of the IMP or (trial related) assessments. At each visit, the questionnaires are to be completed in the same order every time, following the order presented in the protocol.
h. Vital signs (blood pressure, pulse, body temperature) will be measured at t=0.5, 1, 2, 3, 4, 6, 8, 12 and 24 hours after the IMP infusion has been completed.

Table 2 Flow Chart for Laboratory Parameters, for Screening, IMP Administration + Post-Treatment Follow-up (Week -6 to Week 52)

Trial Period	Screening	IMP Administration				Post-treatment Follow-up						Additional Visits [a]
Weeks/Day	W-6	D0 pre IMP [b]	D0 post IMP			D1	D3	W1 to 12 [b, c] (1 visit/W)	W14 to 26 [b, c] (every 2nd W)	W31 to 48 [b, c] (1 visit/month)	W52 [b, c]	
Hours After IMP Infusion Completed			1h	2h	3h	24h						
Visit Window (Days)			±15 minutes			±15 minutes	±1	±2	±3	±5	±5	
Local Laboratory												
Factor IX: one-stage aPTT for factor IX activity for local monitoring		x					x	x	x	x	x	(x) [a]
Factor IX inhibitors (Bethesda assay or Nijmegen modified Bethesda assay) for local monitoring and eligibility check [d]	x	x						x: W6, 12	x: W26		x	(x) [a]
Transaminases (AST/ ALT) for local monitoring [e]	x	x						x	x	x	x	(x) [a]
Central Laboratory												
One-stage aPTT for factor IX activity		x						x	x	x	x	(x) [a]
Chromogenic assay for factor IX activity [f]		x						x	x	x	x	(x) [a]
Factor IX protein concentration		x						x	x	x	x	(x) [a]
Anti-factor IX antibodies		x						x: W6, 12	x: W26		x	(x) [a]
Factor IX inhibitors (Nijmegen modified Bethesda assay) [d]	x	x						x: W6, 12	x: W26		x	(x) [a]
Factor IX Recovery [g]		x										(x) [a]
Total (IgM and IgG) antibodies to AAV5	x	x						x: W1, 2, 3, 4, 5, 6, 12	x: W26		x	(x) [a]
Neutralizing antibodies to AAV5	x	x						x: W1, 2, 3, 4, 5, 6, 12	x: W26		x	(x) [a]
AAV5 capsid-specific T cells		x						x	x	x	x	(x) [a]

Trial Period	**Screening**	**IMP Administration**				**Post-treatment Follow-up**						**Additional Visits** [a]
Weeks/Day	W-6	D0 pre IMP [b]	D0 post IMP			D1	D3	W1 to 12 [b, c] (1 visit/W)	W14 to 26 [b, c] (every 2nd W)	W31 to 48 [b, c] (1 visit/month)	W52 [b, c]	
Hours After IMP Infusion Completed			1h	2h	3h	24h						
Visit Window (Days)			±15 minutes			±15 minutes	±1	±2	±3	±5	±5	
Sampling for vector genome detection:												
- Blood [h]		x	x	x	x	x		x	x	x	x	(x) [a]
- Semen [h]		x						x:W6, 12	x:W16, 26		x	(x) [a]
Inflammatory markers IL-1β, IL-2, IL-6, IFNγ, MCP-1		x						x	x	x	x	(x) [a]
Alpha-fetoprotein											x	(x) [a]
Hematology and coagulation parameters [i]	x	x						x	x	x	x	(x) [a]
Serum chemistry parameters [j]	x	x			x (only CRP)			x	x	x	x	(x) [a]
HIV, viral load, $CD4^+$	x											
HBsAg, HBeAg, HBV DNA and HCV RNA	x											
Factor IX gene sequencing [k]	x											
Blood sample for future research [l]	x	x						x: W12			x	

AAV5: adeno-associated viral vector serotype 5; ALT: alanine aminotransferase; aPTT: activated partial thromboplastin time; AST: aspartate aminotransferase; CRP: C-Reactive Protein; DNA: deoxyribonucleic acid; h: hour; HBeAg: hepatitis B extracellular antigen; HBsAg: hepatitis B surface antigen; HBV: hepatitis B virus; HCV RNA: hepatitis C virus ribonucleic acid; HIV: human immunodeficiency virus; IFNγ: interferon gamma; IgG: immunoglobulin G; IgM: immunoglobulin M; IL: interleukin: IMP: investigational medicinal product; MCP-1: monocyte chemotactic protein-1.

a. An additional visit can be performed at any time for the purpose of conduct of one or more procedures listed in the column "Additional Visits" according to the choice of the Investigator. Hence, "(x)" refers to a procedure that can be performed, if judged relevant by the Investigator.
b. For subjects on routine prophylactic factor IX replacement therapy, these visits should take place on the day that routine prophylactic factor IX replacement treatment is planned to be administered. At these visits, blood sampling will then take place prior to administration of prophylactic factor IX replacement therapy. If a subject uses additional on-demand factor IX replacement treatment, his upcoming study visit may need to be re-scheduled (to the extent the visit window allows) so that the visit takes place at the time the subject resumes his routine prophylaxis schedule. For subjects only using on-demand factor IX replacement therapy (only applicable after AMT-061 administration), his upcoming study visit may need to be re-scheduled (to the extent the visit window allows) so that the visit does not take place within 10 days of any on-demand factor IX product use.
c. Each visit and time point is scheduled in relation to date of IMP administration, not in relation to the date of its previous visit or time point.
d. In case of a positive test for factor IX inhibitors, a re-test should be performed with two weeks to confirm the positive test. The subject should be called in for an additional visit in case no routine visit is scheduled within this two-week timeframe.

e. For ALT level increments of at least 2-fold baseline (i.e., pre-IMP) and/or > upper limit of normal (ULN), by local or central laboratory, the Investigator should contact the Medpace and uniQure Medical Directors to discuss a clinical management plan, on a case-by-case basis, including potential re-tests and/or initiation of corticosteroid treatment. In case of AST level increments > ULN, the Investigator should contact the Medpace and uniQure Medical Directors and a similar discussion should take place.
f. No additional blood will be drawn for the chromogenic assay; the test will be performed on backup samples.
g. At suspicion of factor IX inhibitor, the factor IX recovery assessment should be repeated. Blood samples for factor IX recovery assessment should be drawn just prior to administering the factor IX challenge dose (40 U/kg) and at 30 minutes after the factor IX challenge dose is administered.
h. Sampling for the individual subject and for a specific matrix (i.e., blood or semen) is only to be continued until 3 consecutive negative samples have been detected for the subject for that particular type of matrix. The sampling schedule (frequency) may be increased as agreed between the Investigator and subject following the notification of a first negative result on blood and/or semen, expediting the opportunity to reach three consecutive negative samples on the specific matrix. Based on the wish of the subject, semen samples can be collected at home prior to attending the visit (at the visit day or at the day before the visit day). Also, the frequency of semen sampling may be reduced (to be agreed between Investigator and subject) as long as the subject uses a condom during sexual intercourse until 3 consecutive negative samples have been detected.
i. For hematology: Hemoglobin, hematocrit, platelet count, red blood cells, white blood cells with differential count (all expressed in % as well as in absolute numbers); for coagulation: aPTT and prothrombin time (or INR [International Normalized Ratio])).
j. Serum electrolytes (sodium, potassium), creatinine, creatine kinase, gamma-glutamyltransferase (γGT), AST, ALT, alkaline phosphatase (ALP), CRP, albumin, total bilirubin, glucose (non-fasting).
k. Only if factor IX gene mutation information is not available and if separate informed consent for factor IX gene sequencing analysis is given by the subject. Preferably at the screening visit, but otherwise at a later time point during the subject's trial participation.
l. Only if separate informed consent is given by the subject.

Table 3 Flow Chart for Efficacy and Safety Evaluation, for Long-Term Follow-up (up to Month 60)

Trial Period	Long-Term Follow-up								Additional Visits [a]
Months	18 [b, c]	24 [b, c]	30 [b, c]	36 [b, c]	42 [b, c]	48 [b, c]	54 [b, c]	60 [b, c]	
Visit Window (Weeks)	±2	±2	±2	±2	±2	±2	±2	±2	
Concomitant medication	Continuous								(x) [a]
Collection of bleeding and factor IX use data [d]	x	x	x	x	x	x	x	x	(x) [a]
Physical examination	x	x	x	x	x	x	x	x	(x) [a]
Blood & semen sampling (see Table 4)	x	x	x	x	x	x	x	x	(x) [a]
CCI		x		x		x		x	
CCI		x		x		x		x	
CCI		x		x		x		x	
CCI		x		x		x		x	
CCI		x		x		x		x	
CCI		x		x		x		x	
Abdominal ultrasound [f]			x	x	x	x	x	x	(x) [a]
Adverse events [g]	Continuous								

CCI ; IMP: investigational medicinal product; CCI

a. An additional visit can be performed at any time for the purpose of conduct of one or more procedures listed in the column "Additional Visits" according to the choice of the Investigator. Hence, "(x)" refers to a procedure that can be performed, if judged relevant by the Investigator.

b. Each visit and time point is scheduled in relation to date of IMP administration, not in relation to the date of its previous visit or time point.

c. For subjects on routine prophylactic factor IX replacement therapy, these visits should take place on the day that routine prophylactic factor IX replacement treatment is planned to be administered. At these visits, blood sampling will then take place prior to administration of prophylactic factor IX replacement therapy. If a subject uses additional on-demand factor IX replacement treatment, his upcoming study visit may need to be re-scheduled (to the extent the visit window allows) so that the visit takes place at the time the subject resumes his routine prophylaxis schedule. For subjects only using on-demand factor IX replacement therapy (only applicable after AMT-061 administration), his upcoming study visit may need to be re-scheduled (to the extent the visit window allows) so that the visit does not take place within 10 days of any on-demand factor IX product use.

d. Subjects will record bleeding episodes and factor IX replacement therapy use in their study-specific paper long-term follow-up bleed diary and paper long-term follow-up factor IX use diary, which they will bring to every study visit. Site staff will collect the information that is new since the previous visit in the paper diaries. In between study visits, subjects should contact the site staff immediately in case of an experienced bleed and/or factor IX use in addition to completing the questions/information requested on the paper diaries.

e. It is recommended that the CCI (CCI) are completed by the subject before he is seen by the physician for interview and before other (trial specific) assessments are performed, as well as before the administration of IMP, in order for the answers not to be influenced by the information given by the physician, by the administration of IMP, or by early side-effects of the IMP or (trial related) assessments. During the screening visit, the CCI are completed after ICF signature, but before other (trial specific) assessments are performed. At each visit, the questionnaires are to be completed in the same order every time, following the order presented in the protocol. The questionnaires are to be completed under site supervision at the site. For those subjects where the Month 24, Month 36, Month 48, and/or Month 60 visits are impacted by COVID-19, the questionnaires may be conducted at the site within the following window: up to -1 month prior to the target visit, up to +2 months after the target visit, and at least 4 months after the last assessment. Adjustments to the visit timing within this window are to be documented.

f. For those subjects where Month 30, Month 36, Month 42, Month 48, Month 54, and/or Month 60 visits are impacted by COVID-19, abdominal ultrasounds may be conducted within the following window: up to -1 month prior to the target visit and up to +1 month after the target visit. Adjustments to the visit timing within this window are to be documented.

g. After week 52, the visit frequency is every 26 weeks (6 months). In this long-term follow-up phase there should be at least quarterly contact between the site staff and subject to monitor the occurrence of adverse events.

Table 4 Flow Chart for Laboratory Parameters, for Long-Term Follow-up (up to Month 60)

Trial Period	Long-Term Follow-up								Additional Visits [a]
Months	18 [b, c]	24 [b, c]	30 [b, c]	36 [b, c]	42 [b, c]	48 [bb, c]	54 [b, c]	60 [b, c]	
Visit Window (Weeks)	±2	±2	±2	±2	±2	±2	±2	±2	
Local Laboratory									
Factor IX: one-stage aPTT for factor IX activity for local monitoring			x	x	x	x	x	x	(x) [a]
Transaminases (AST/ ALT) for local monitoring [d]			x	x	x	x	x	x	(x) [a]
Central Laboratory									
One-stage aPTT for factor IX activity	x	x	x	x	x	x	x	x	(x) [a]
Chromogenic assay for factor IX activity [e]	x	x	x	x	x	x	x	x	(x) [a]
Factor IX protein concentration	x	x	x	x	x	x	x	x	(x) [a]
Anti-factor IX antibodies		x		x		x		x	(x) [a]
Factor IX inhibitors (Nijmegen modified Bethesda assay) [f]		x		x		x		x	(x) [a]
factor IX Recovery [g]									(x) [a]
Total (IgM and IgG) antibodies to AAV5		x		x		x		x	(x) [a]
Neutralizing antibodies to AAV5		x		x		x		x	(x) [a]
Sampling for vector genome detection:									
- Blood [h]	x	x	x	x	x	x	x	x	(x) [a]
- Semen [h]	x	x	x	x	x	x	x	x	(x) [a]
Alpha-fetoprotein	x	x	x	x	x	x	x	x	(x) [a]
Hematology and coagulation parameters [i]	x	x	x	x	x	x	x	x	(x) [a]
Serum chemistry parameters [j]	x	x	x	x	x	x	x	x	(x) [a]

AAV5: adeno-associated viral vector serotype 5; ALT: alanine aminotransferase; aPTT: activated partial thromboplastin time; AST: aspartate aminotransferase; IgG: immunoglobulin G; IgM: immunoglobulin M; IMP: investigational medicinal product.

a. An additional visit can be performed at any time for the purpose of conduct of one or more procedures listed in the column "Additional Visits" according to the choice of the Investigator. Hence, "(x)" refers to a procedure that can be performed, if judged relevant by the Investigator.

b. Each visit and time point is scheduled in relation to date of IMP administration, not in relation to the date of its previous visit or time point.

c. For subjects on routine prophylactic factor IX replacement therapy, these visits should take place on the day that routine prophylactic factor IX replacement treatment is planned to be administered. At these visits, blood sampling will then take place prior to administration of prophylactic factor IX replacement therapy. If a subject uses additional on-demand factor IX replacement treatment, his upcoming study visit may need to be re-scheduled (to the extent the visit window allows) so that the visit takes place at the time the subject resumes his routine prophylaxis schedule. For subjects only using on-demand factor IX replacement therapy (only applicable after AMT-061

administration), his upcoming study visit may need to be re-scheduled (to the extent the visit window allows) so that the visit does not take place within 10 days of any on-demand factor IX product use.

d. For ALT level increments of at least 2-fold baseline (i.e., pre-IMP) and/or > upper limit of normal (ULN), by local or central laboratory, the Investigator should contact the Medpace and uniQure Medical Directors to discuss a clinical management plan, on a case-by-case basis, including potential re-tests and/or initiation of corticosteroid treatment. In case of AST level increments > ULN, the Investigator should contact the Medpace and uniQure Medical Directors and a similar discussion should take place.
e. No additional blood will be drawn for the chromogenic assay; the test will be performed on backup samples.
f. In case of a positive test for factor IX inhibitors, a re-test should be performed with two weeks to confirm the positive test. The subject should be called in for an additional visit in case no routine visit is scheduled within this two-week timeframe.
g. At suspicion of factor IX inhibitor, the factor IX recovery assessment should be repeated. Blood samples for factor IX recovery assessment should be drawn just prior to administering the factor IX challenge dose (40 U/kg) and at 30 minutes after the factor IX challenge dose is administered.
h. Sampling for the individual subject and for a specific matrix (i.e., blood or semen) is only to be continued until 3 consecutive negative samples have been detected for the subject for that particular type of matrix. The sampling schedule (frequency) may be increased as agreed between the Investigator and subject following the notification of a first negative result on blood and/or semen, expediting the opportunity to reach three consecutive negative samples on the specific matrix. Based on the wish of the subject, semen samples can be collected at home prior to attending the visit (at the visit day or at the day before the visit day). Also, the frequency of semen sampling may be reduced (to be agreed between Investigator and subject) as long as the subject uses a condom during sexual intercourse until 3 consecutive negative samples have been detected. Semen samples are only to be collected as applicable.
i. For hematology: hemoglobin, hematocrit, platelet count, red blood cells, white blood cells with differential count (all expressed in % as well as in absolute numbers); for coagulation: aPTT and prothrombin time (or INR [International Normalized Ratio]).
j. Serum electrolytes (sodium, potassium), creatinine, creatine kinase, gamma-glutamyltransferase (γGT), aspartate aminotransferase (AST), alanine transaminase (ALT), alkaline phosphatase (ALP), C-Reactive Protein (CRP), albumin, total bilirubin, glucose (non-fasting).